

HEART DISEASE PREDICTION USING MACHINE LEARNING

- **Parth Navadiya-22202538**
- **Narender Kumar**
 - B.Sc. Artificial Intelligence
 - Deggendorf Institute of Technology

Presentation Outline

Motivation

Dataset Overview

Exploratory Data Analysis

Machine Learning Evaluation Pipeline

Model Development

Experimental Results

Discussion

Conclusion



Motivation

Why Heart Disease Prediction?

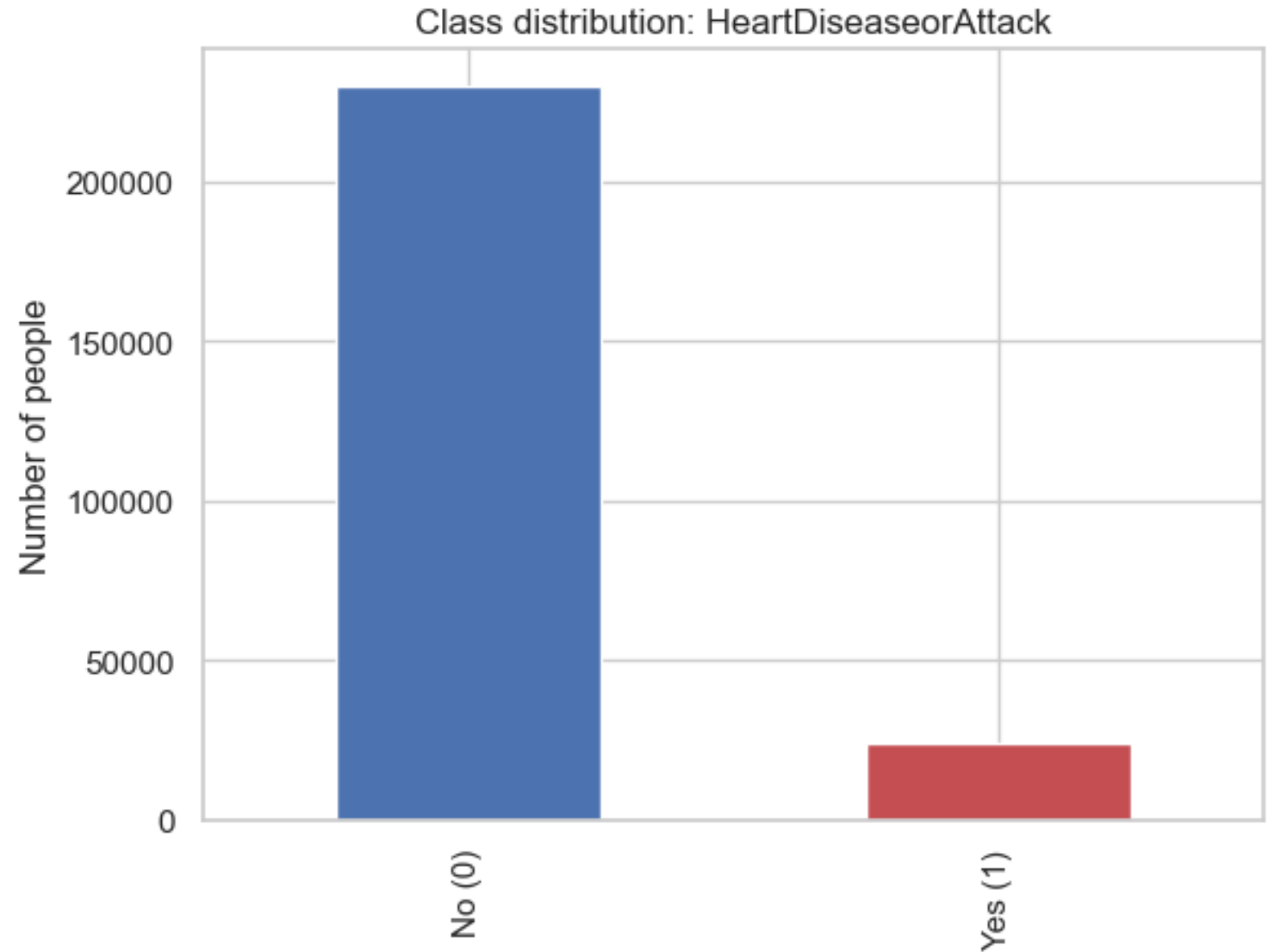
- Leading cause of death worldwide
- Early detection improves treatment
- Machine learning can identify patterns from health survey data.
- Proper evaluation is essential to select a reliable prediction model.

DATASET OVERVIEW

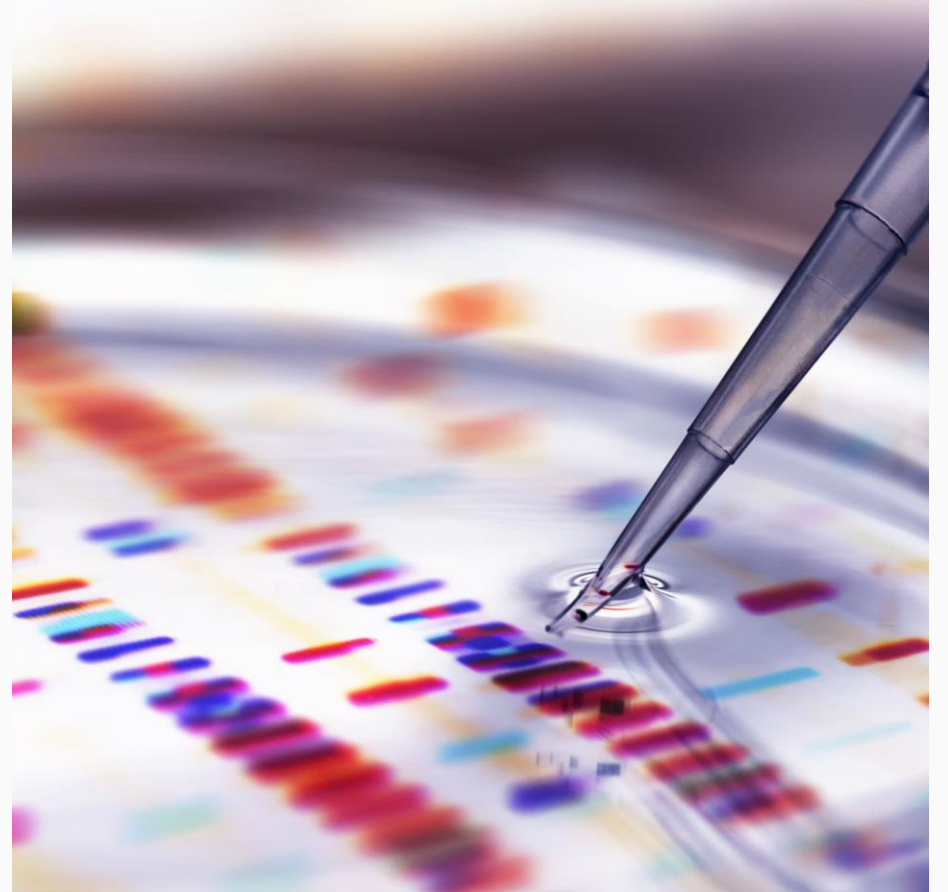
Property	Description
Dataset	BRFSS 2015
Source	CDC Behavioral Risk Factor Surveillance System
Samples	253,680
Features	21 Health Indicators
Target	HeartDiseaseorAttack
Task	Binary Classification

DATASET CHARACTERISTICS

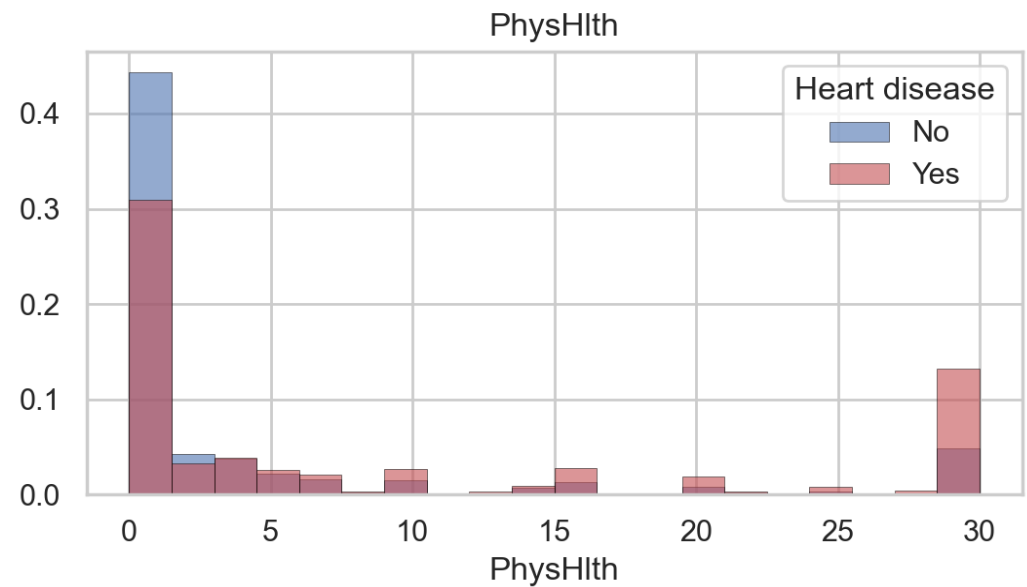
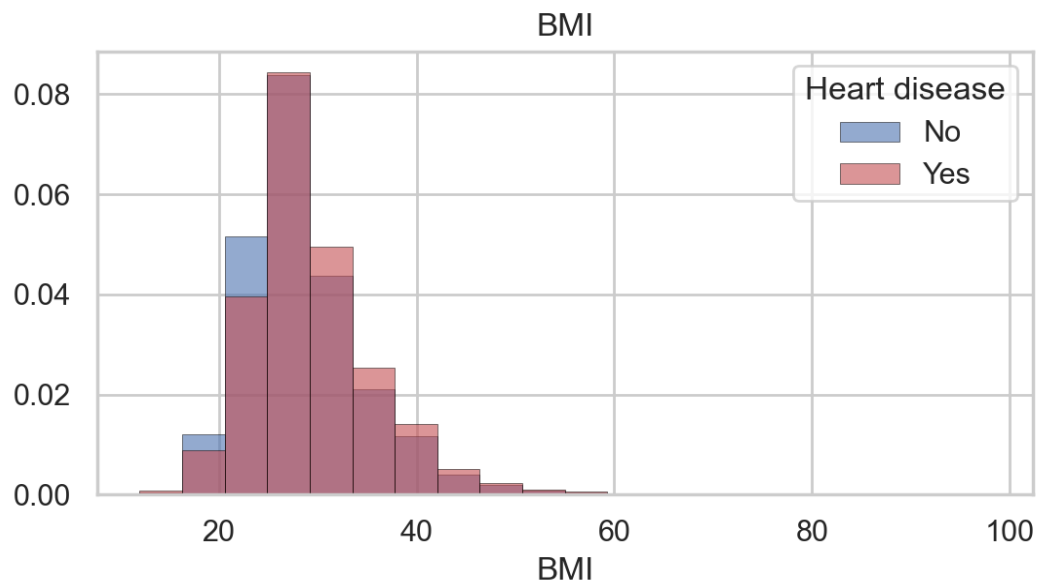
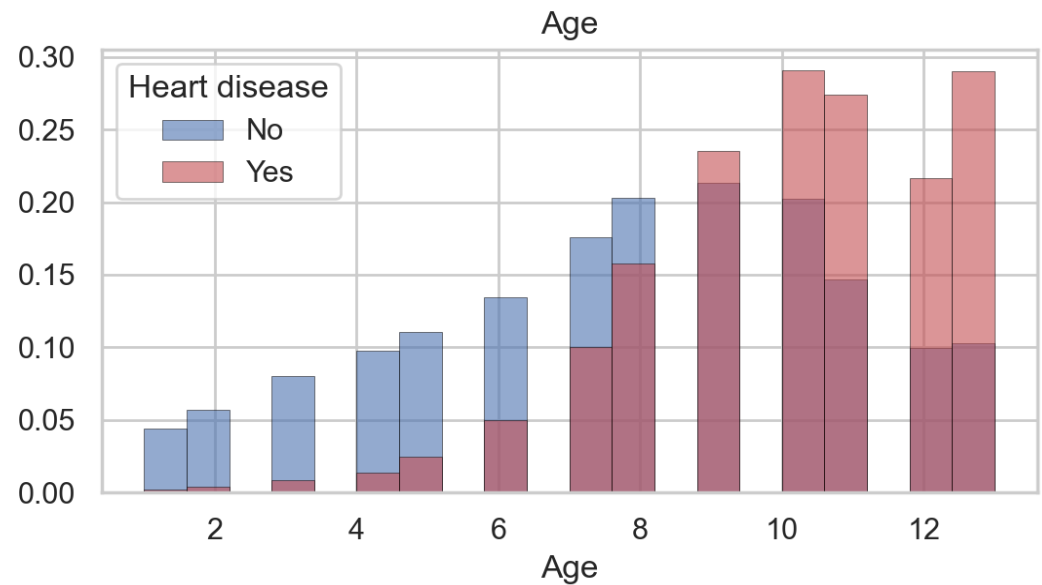
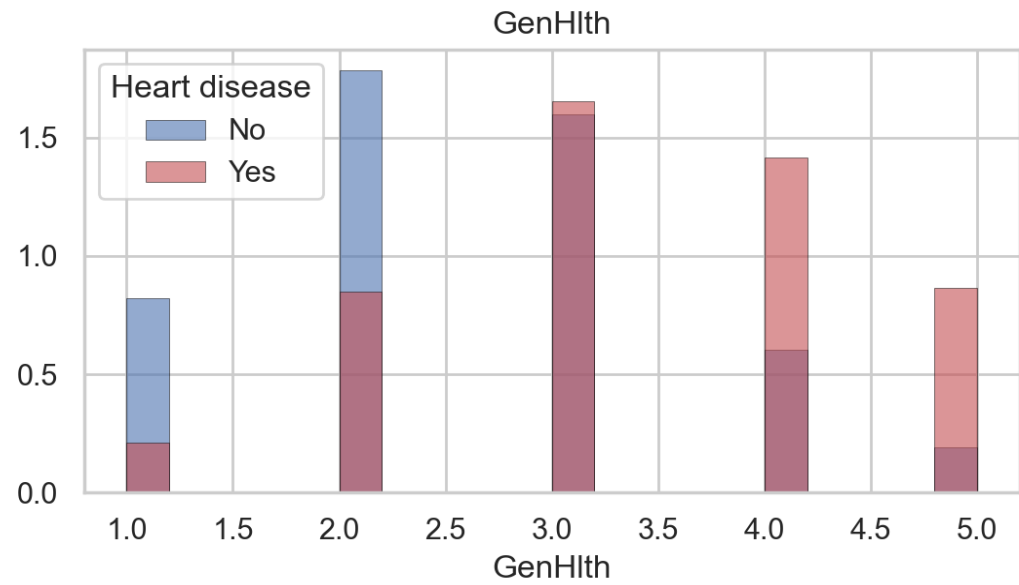
- Majority class = Healthy
- Minority class = Heart Disease
- Strong class imbalance
- 253,680 responses (229,781 after removing duplicates)
- 21 numeric features + 1 binary target

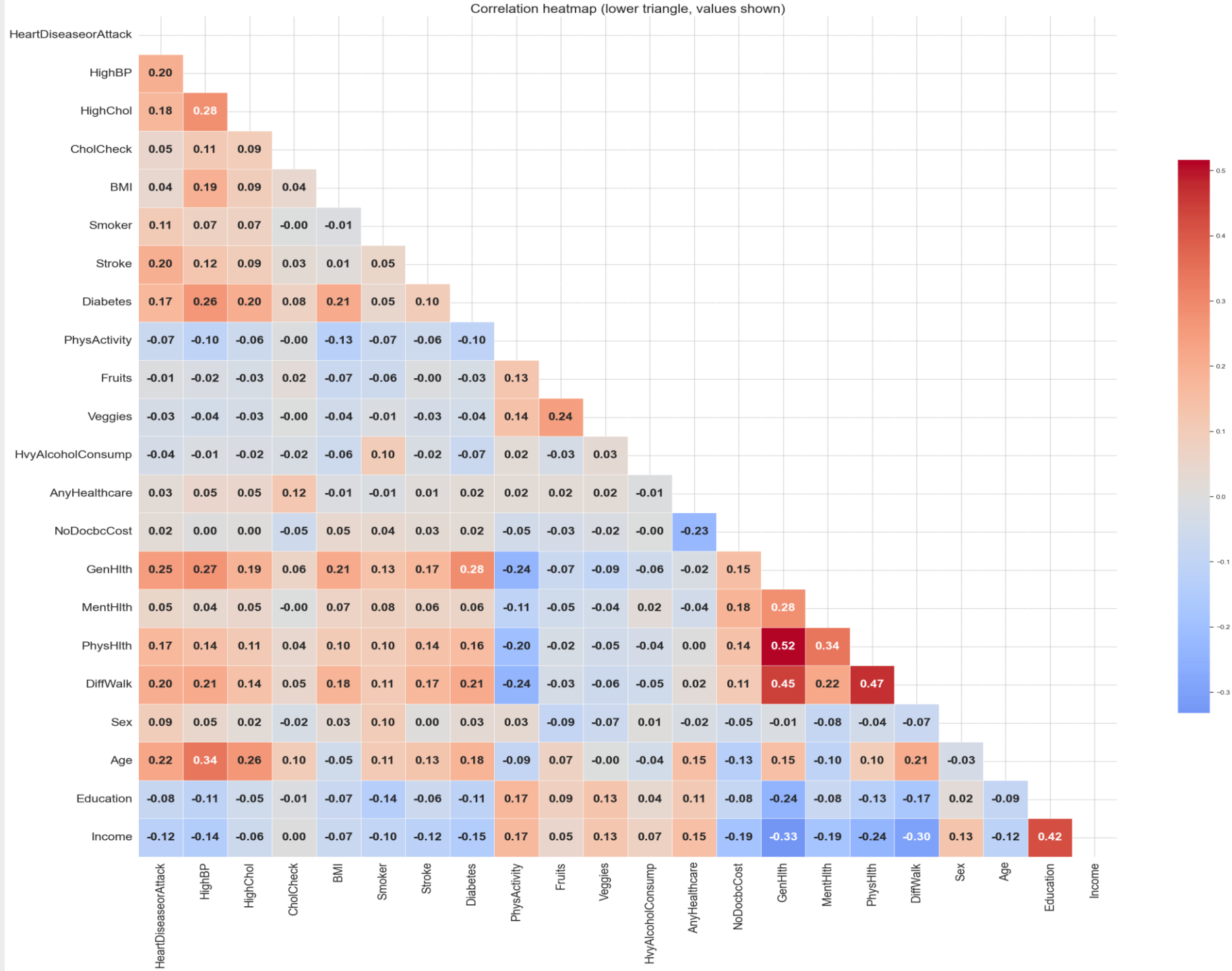


EXPLORATORY DATA ANALYSIS



Feature distribution by class (red = has heart disease)





Correlation Heatmap

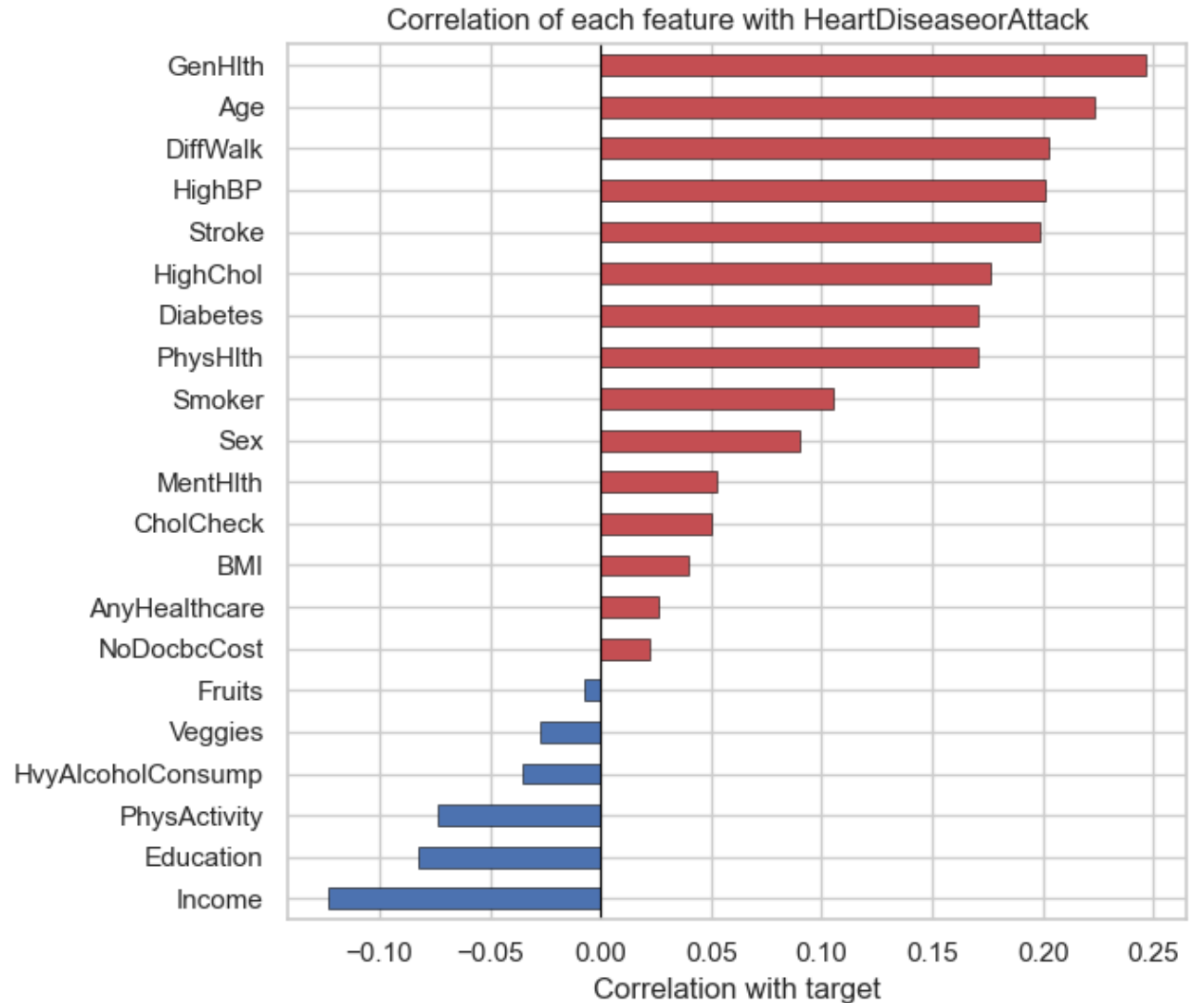
Key Observations:

- Most features have **weak to moderate correlations**, indicating **low multicollinearity**.
- **General Health** and **Physical Health** show the strongest positive correlation ($r = 0.52$).
- **Age, High Blood Pressure, and Stroke** are positively associated with **Heart Disease**.
- No highly correlated features were found, so **Forward Feature Selection** was used to identify the most important features.

CORRELATION WITH TARGET VARIABLE

Most Influential Features:

- Age
- General Health
- High Blood Pressure
- High Cholesterol
- Difficulty Walking



Forward Feature Selection

Purpose:

- Identify a smaller subset of informative features while maintaining predictive performance.

Selected Features:

- CholCheck, BMI, Stroke, PhysActivity, AnyHealthcare, NoDocbcCost, MentHlth, Education

Why did we do this?

- Reduce dimensionality.
- Remove less informative features.
- Compare models trained on all features versus selected features.

Machine Learning Evaluation Pipeline

Dataset

Data Cleaning

Exploratory Data Analysis

Train/Test Split

Feature Scaling

Forward Feature Selection

Model Training

Hyperparameter Optimization

Model Evaluation

Model Comparison

Experimental Setup

Why?

- Same experimental setup for all models.
- Hyperparameter tuning performed **only on the training data**.
- Final evaluation performed on the unseen test set.

Step	Method
Train/Test Split	70% Training / 30% Testing
Feature Scaling	StandardScaler (kNN & Linear SVM)
Feature Selection	Forward Feature Selection
Hyperparameter Optimization	GridSearchCV
Cross Validation	3-Fold Cross Validation
Evaluation Metrics	Accuracy, Balanced Accuracy, Sensitivity, Specificity, ROC-AUC

Models Evaluated

Hyperparameter Optimization

- **Method:** GridSearchCV
- **Validation:** 3-Fold Cross Validation
- **Selection:** Best average validation score

Purpose

- Finds the optimal hyperparameters.
- Improves model performance.
- Reduces manual tuning.

Model	Role	Hyperparameters
kNN	Baseline	Number of Neighbors (k)
Random Forest	Advanced	n_estimators, max_depth, max_features
Linear SVM	Advanced	C (Regularization Parameter)

Baseline Model – k-Nearest Neighbors (kNN)

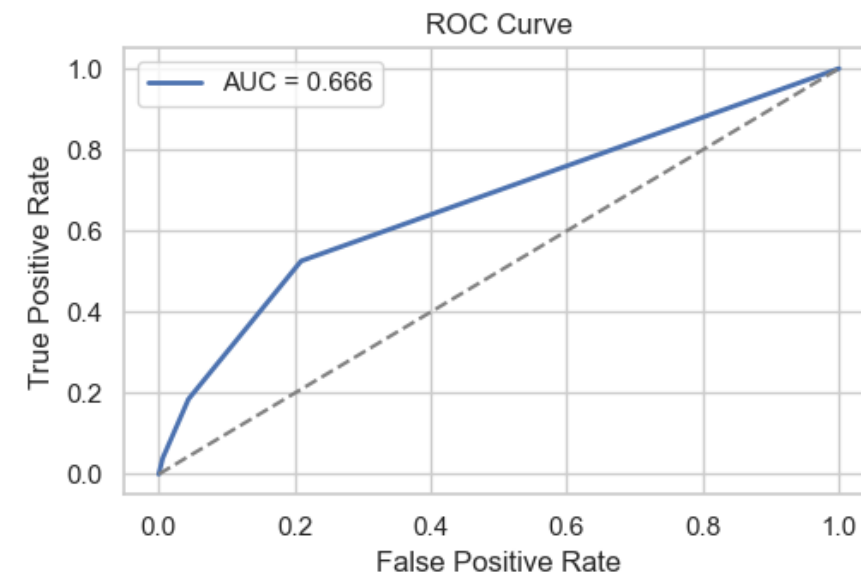
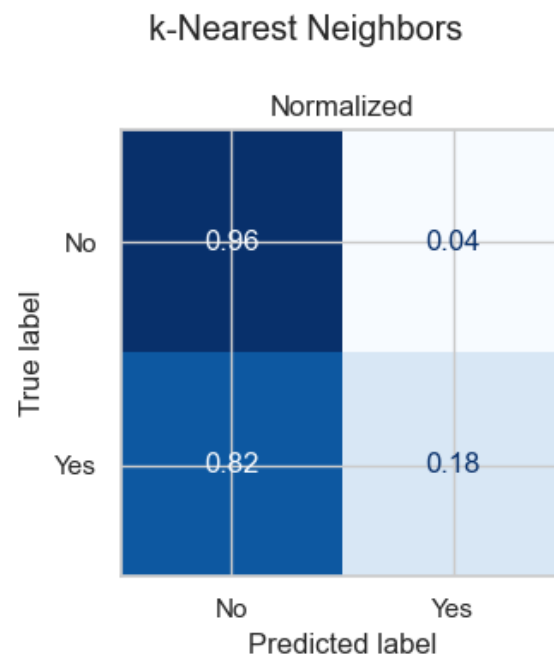
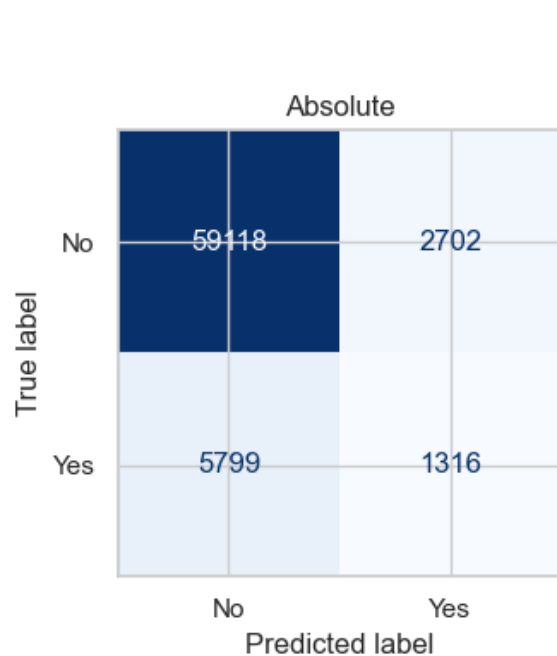
Configuration:

- StandardScaler
- Selected Features
- Hyperparameter optimized using 3-Fold GridSearchCV

Results:

Metric	Value
Accuracy	87.7%
Balanced Accuracy	57.1%
Sensitivity	18.5%
Specificity	95.6%
ROC-AUC	0.666

- Baseline model achieved good overall accuracy but struggled to detect positive heart disease cases.



Key Observation:

- Correctly classified most healthy individuals but missed many heart disease cases.
- Low sensitivity (**18.5%**) indicates poor detection of the minority class.
- ROC-AUC (**0.666**) shows only **fair** classification performance.

Random Forest

Configuration:

- All Features and Selected Features
- GridSearchCV (3-Fold CV)
- Scoring: Balanced Accuracy

Parameter Search

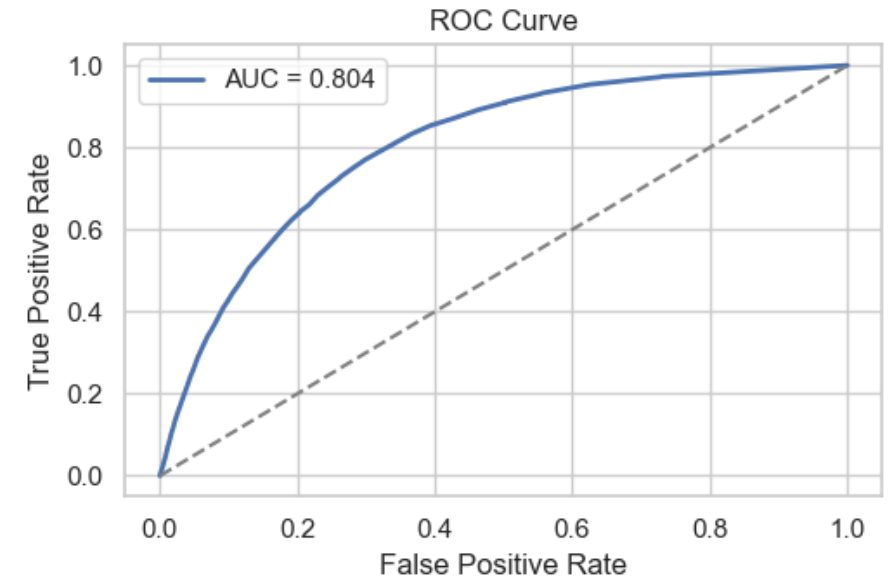
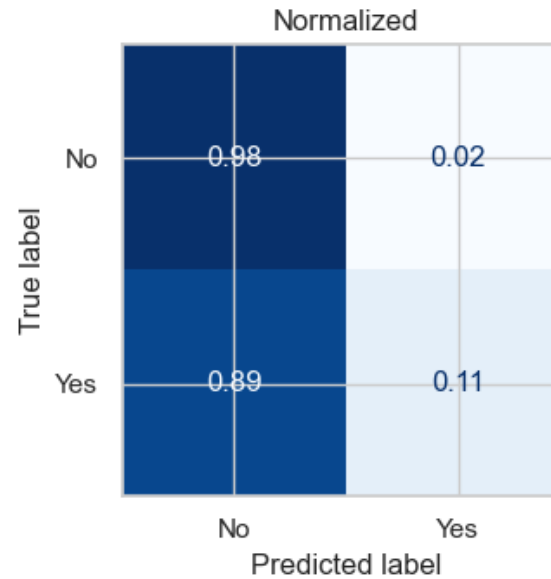
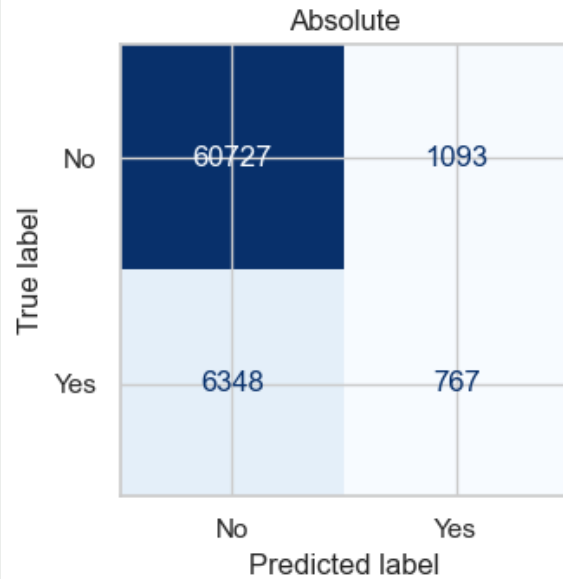
- n_estimators: 100, 200
- max_depth: None, 10
- max_features: sqrt, log2

Best: 100 | None | sqrt

Metric	All Features	Selected Features
Accuracy	89.2%	89.2%
Balanced Accuracy	54.5%	52.0%
Sensitivity	10.8%	5.2%
Specificity	98.2%	98.8%
ROC-AUC	0.804	0.622

- Feature selection reduced sensitivity and ROC-AUC while accuracy remained unchanged.
- The optimized model with **all features** performed better overall.

Random Forest (Optimized)



*Optimized with all features.

Key Observations:

- Higher overall accuracy than kNN with better class separation.
- Majority class was classified well, but sensitivity remained low (**10.8%**).
- ROC-AUC (**0.804**) indicates **good** discriminative performance.

Linear SVM

Configuration:

- Features: All Features and Selected Features
- Scaling: StandardScaler
- Tuning: GridSearchCV (3-Fold CV)
- Scoring: Balanced Accuracy

Parameter Search:

- C: 0.01, 0.1, 1, 10, 100

Best Parameter:

- C = 1

Metric	All Features	Selected Features
Accuracy	89.9%	89.7%
Balanced Accuracy	52.3%	50.0%
Sensitivity	5.0%	0.1%
Specificity	99.6%	100.0%
ROC-AUC	0.836	0.651

- Highest overall accuracy.
- Lowest sensitivity.
- Selected features reduced minority detection.

Linear SVM (Optimized)

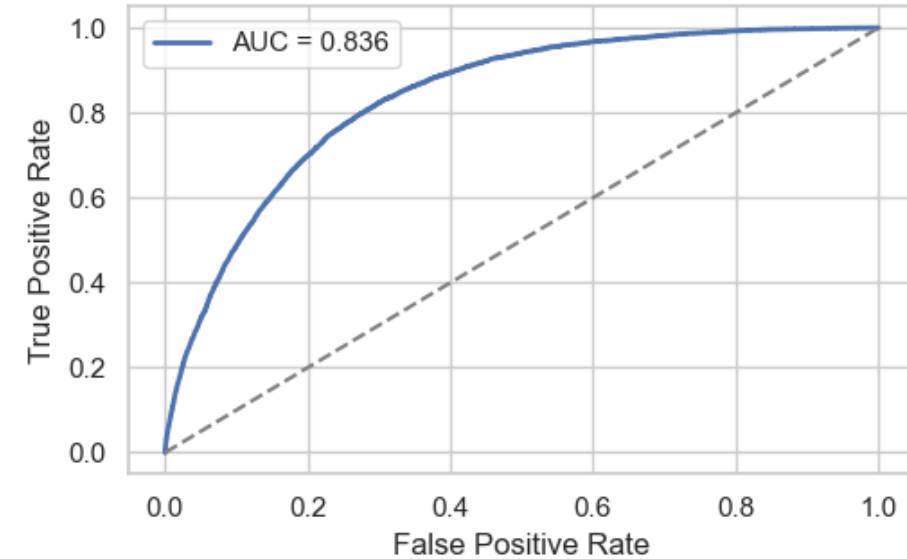
Absolute

True label	Predicted label	
	No	Yes
No	61595	225
Yes	6756	359

Normalized

True label	Predicted label	
	No	Yes
No	1.00	0.00
Yes	0.95	0.05

ROC Curve



*Optimized with all features.

Key Observation:

- Achieved the highest overall accuracy (**89.9%**).
- Correctly classified most healthy individuals but detected very few heart disease cases (**Sensitivity: 5.0%**).
- ROC-AUC (**0.836**) indicates good discrimination despite low sensitivity.

Initial Model Comparison

Observation:

- All models achieved **high accuracy (>87%)**.
- Sensitivity remained very low.
- Models favored the majority (healthy) class.

Model	Accuracy	Balanced Accuracy	Sensitivity	ROC-AUC
kNN	87.7%	57.1%	18.5%	0.666
Random Forest	89.2%	54.5%	10.8%	0.804
Linear SVM	89.9%	52.3%	5.0%	0.836

Why Was Sensitivity Low?

Observation

- Majority class dominated the dataset.
- Models learned the majority class.
- Heart disease cases were frequently misclassified.

Key Takeaway:

- Accuracy alone was misleading because the models rarely predicted the minority class correctly.



Handling Class Imbalance

Approach:

- Applied `class_weight="balanced"`
- Retrained Random Forest
- Retrained Linear SVM

Objective:

- Increase the importance of the minority class during training and improve heart disease detection.

Model	Accuracy	Balanced Accuracy	Sensitivity	Specificity	ROC-AUC
Random Forest (Balanced)	86.8%	63.4%	34.0%	92.8%	0.808
Linear SVM (Balanced)	73.6%	76.1%	79.2%	73.0%	0.837

Key Observations:

- Sensitivity improved substantially.
- Balanced Accuracy increased.
- Small reduction in overall Accuracy.

Final Model Comparison

Model	Accuracy	Balanced Accuracy	Sensitivity	ROC-AUC
kNN (Baseline)	87.7%	57.1%	18.5%	0.666
Random Forest (Default)	89.2%	54.5%	10.8%	0.804
Random Forest (Balanced)	86.8%	63.4%	34.0%	0.808
Linear SVM (Default)	89.9%	52.3%	5.0%	0.836
Linear SVM (Balanced)	73.6%	76.1%	79.2%	0.837

- Highest Accuracy → **Linear SVM (Default)**
- Best Medical Performance → **Linear SVM (Balanced)**

Key Findings

- The dataset was highly imbalanced, making accuracy alone misleading.
- Hyperparameter optimization had minimal impact on model performance.
- Applying `class_weight="balanced"` significantly improved minority class detection.
- Balanced Linear SVM achieved the best trade-off between sensitivity and overall performance.

Conclusion

- Evaluated one baseline and two advanced classifiers.
- Compared all features and selected features.
- Addressed class imbalance using class weights.
- Balanced Linear SVM was selected as the final model because it provided the best balance between detecting heart disease cases (high sensitivity) and maintaining overall classification performance.

THANK YOU